

Homework 9

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1. Show that a linear functional T on a vector space X is uniquely determined by its values on a Hamel basis for X . In other words, prove

- Given a Hamel basis S , any function $f : S \rightarrow \mathbb{R}$ defines a linear functional $T : X \rightarrow \mathbb{R}$ with the property that $T(v) = f(v)$ for all $v \in S$.
- If T and T' are two linear functionals on X that are equal on S , then $T = T'$.

Proof. Given a Hamel basis S for X and a function $f : S \rightarrow \mathbb{R}$, we define $T : X \rightarrow \mathbb{R}$ as follows. Given $x \in X$, we can write $x = c_1 s_1 + \cdots + c_n s_n$ where each $s_i \in S$. Then

$$Tx := c_1 f(s_1) + \cdots + c_n f(s_n).$$

This is linear by construction, and it also guarantees by definition that $T(s) = f(s)$ for each $s \in S$.

Now if T and T' are two linear functionals on X which agree on S , then for any $x \in X$, write $x = c_1 s_1 + \cdots + c_n s_n$ where each $s_i \in S$, and see that

$$\begin{aligned} T(x) &= T(c_1 s_1) + \cdots + T(c_n s_n) && \text{(by additivity)} \\ &= c_1 T(s_1) + \cdots + c_n T(s_n) && \text{(by homogeneity)} \\ &= c_1 T'(s_1) + \cdots + c_n T'(s_n) \\ &= T'(c_1 s_1) + \cdots + T'(c_n s_n) \\ &= T'(c_1 s_1 + \cdots + c_n s_n) \\ &= T'(x). \end{aligned}$$

Thus $T = T'$. □

2. Show that any infinite-dimensional normed vector space X must have at least one unbounded linear functional. Hint: use the fact that X has a Hamel basis.

Proof. Given a Hamel basis S for X , choose an injective function $g : \mathbb{Z}^+ \rightarrow S$ and let $S_c = \text{range}(g)$. We can define the function $f : S \rightarrow \mathbb{R}$ by

$$f(s) = \begin{cases} g^{-1}(s) \cdot \|s\| & s \in S_c \\ 0 & \text{otherwise.} \end{cases}$$

Then f determines a linear functional T on X by the previous exercise, and f is unbounded since when s varies across S_c ,

$$\begin{aligned}\frac{|Ts|}{||s||} &= \frac{|g^{-1}(s) \cdot (||s||)|}{||s||} \\ &= \frac{||s|| \cdot |g^{-1}(s)|}{||s||} \\ &= |g^{-1}(s)|,\end{aligned}$$

and $g^{-1}(s)$ will vary across $\mathbb{Z}^+$.

□